



Technical Writing:

Linking is DITA XML

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What Will You Learn?

- What is linking
- Types of links you can create in DITA
- How to manually create a link to a topic, a web page, a resource or a topic section in DITA
- How to automatically produce links between the topics of your content using collection-type attribute in DITA

What is Linking?

As per definition, linking stands for “connecting or joining something to something else.”



Types of Links in DITA

- ▶ In technical writing context in DITA, linking stands for creating a link that **connects a topic** from the documentation **to**:
 - ❖ Web page
 - ❖ A topic in the same map or in another map
 - ❖ A resource
 - ❖ A place inside the topic itself
- ▶ Links can be created **manually** or **automatically**

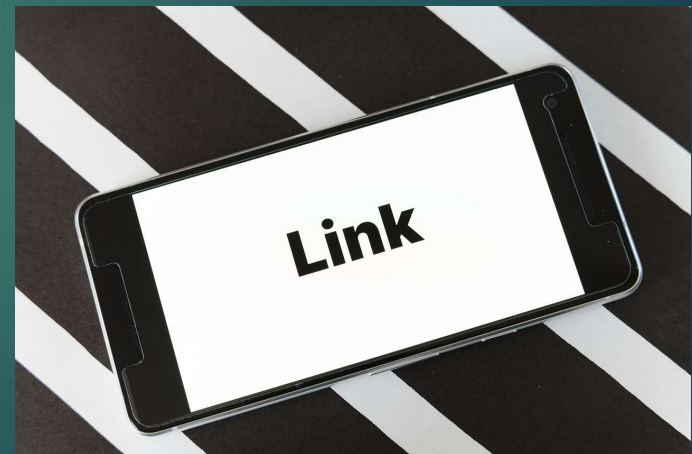
Manually Created Links

- ▶ You can create manually a link to:
 - ▶ Web site.
 - ▶ Another DITA Topic.
 - ▶ A resource to download.
 - ▶ Place in the same document.

Web Link in DITA XML

e**X**ternal **RE**ference -> xref

```
<xref href="http://www.jpdocu.com" format="html" scope="external"/>
```



Link to Another Topic

e**X**ternal **RE**ference -> xref

```
<xref href="topic_3.dita"/>
```

Create a Link to a Resource

e**X**ternal **RE**ference -> xref

For example, to download a JPG file:

```
<xref href="consultancy.JPG" format="JPG"/>
```

Demo and Exercise

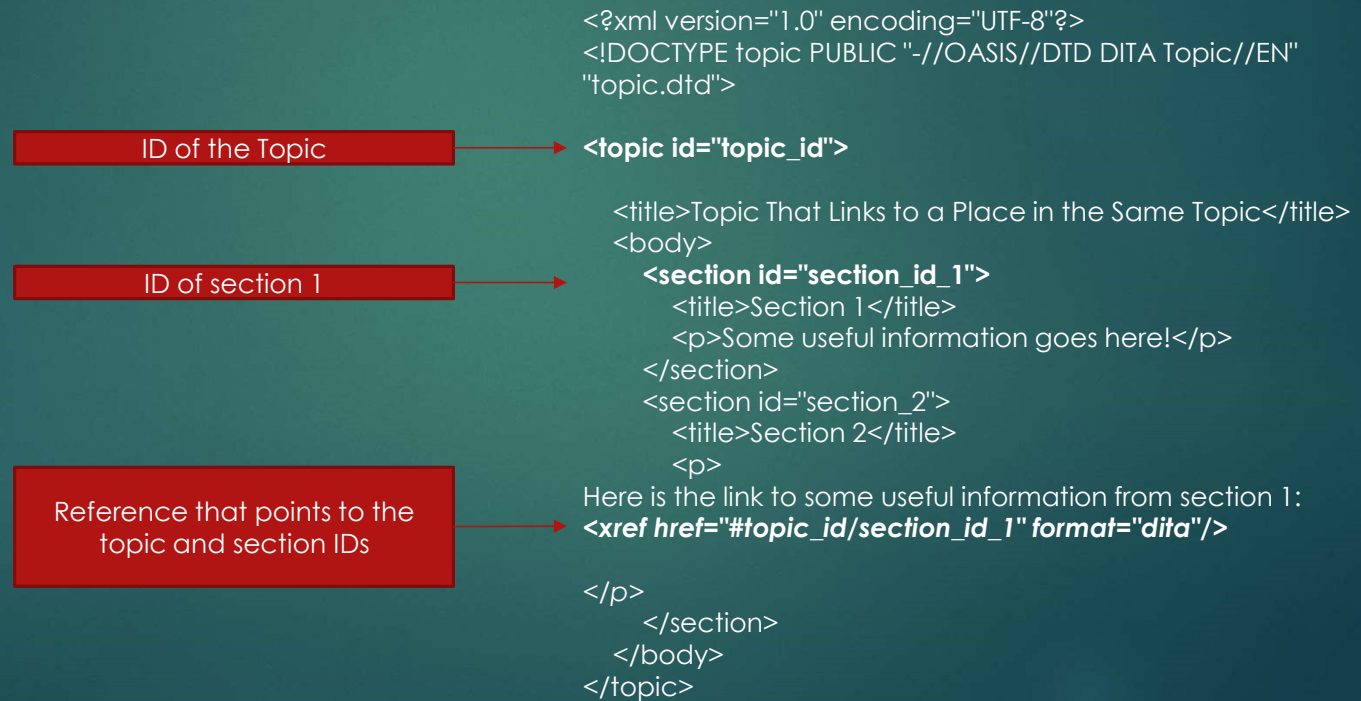
- ▶ Create a link to a web page on the Internet.
- ▶ Create a link to a topic in your documentation.
- ▶ Create a link to a resource to download.

Link a Location in the Same Topic

1. Create an ID for the target location. Most often, you will use <section> topic element to do that.
2. Create a new reference to the topic (itself). Point to the ID of the section.
3. Sample DITA XML source follows.



Link a Location in the Same Topic – Sample DITA XML Source



Demo and Exercise

- ▶ Create a dedicated topic.
- ▶ Create 2 sections inside it.
- ▶ Assign IDs to these sections.
- ▶ Create a paragraph that links to one of the sections.



Automated Links

GENERATE LINKS AUTOMATICALLY, UPON THE OUTPUT GENERATION

Automatic Production of Links

- ▶ You can have links between:
 - ▶ Parent-child topics
 - ▶ Sibling topics
 - ▶ Unordered links



Automatic Generation of Links

To trigger the automated link generation on your entire deliverable, define a value for the *collection-type* attribute on your DITA map.



DITA Maps Manager

Root map: <Current map>

my_dita_map.ditamap

- My DITA Map Title
 - Topic 1
 - Topic 2
 - Topic 3
 - Topic 3.1
 - Topic 3.2
 - Topic 3.3

Edit Properties

Target

Keys

Attributes

Metadata

Profiling

Title: My DITA Map Title

Set to: ☐ Attribute ☒ Element ☐

All attributes:	Attribute	Value
	collection-type	unordered
	anchorref	
	audience	
	base	
	cascade	

Collection Type “choice”

- ▶ Parent-child links
- ▶ Auto generate links to child topics, if present.
- ▶ Suitable for HTML output.



Links to the child topics:

Topic 3

<<Text of this topic is here>>

[Topic 3.1](#)

[Topic 3.2](#)

[Topic 3.3](#)

Link back to the parent topic:

Topic 3.1

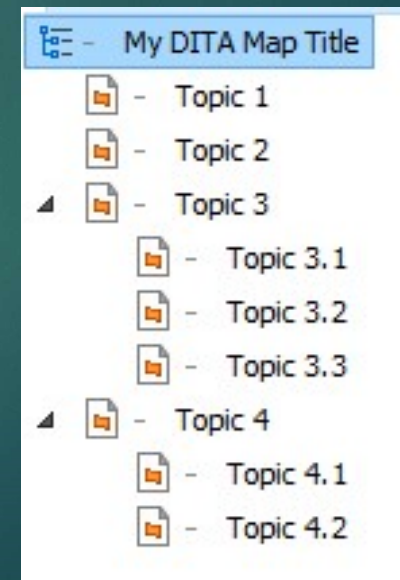
<<Text of this topic is here>>

Parent topic: [Topic 3](#)

Exercise – Step 1

Create a sample DITA map with topics in hierarchical and non-hierarchical relation. We shall use it generate outputs with different types of links:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE map PUBLIC "-//OASIS//DTD DITA Map//EN" "map.dtd">
<map>
  <title>My DITA Map Title</title>
  <topicref href="topic_1.dita"/>
  <topicref href="topic_2.dita"/>
  <topicref href="topic_3.dita">
    <topicref href="topic_3_1.dita"/>
    <topicref href="topic_3_2.dita"/>
    <topicref href="topic_3_3.dita"/>
  </topicref>
  <topicref href="topic_4.dita">
    <topicref href="topic_4_1.dita"/>
    <topicref href="topic_4_2.dita"/>
  </topicref>
</map>
```

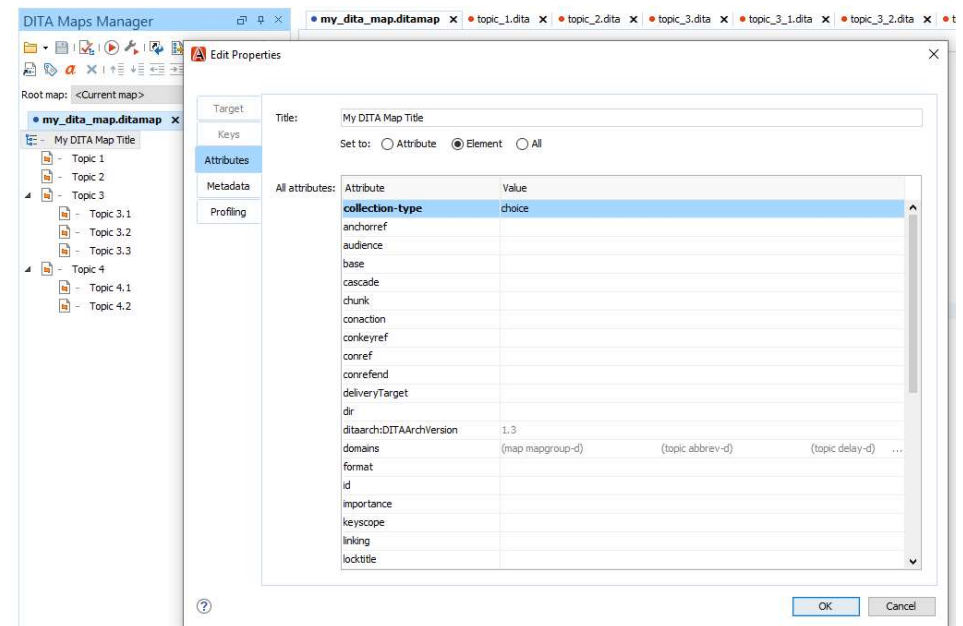


Exercise – Step 2

Set the collection-type attribute of the DITA map to “choice”.

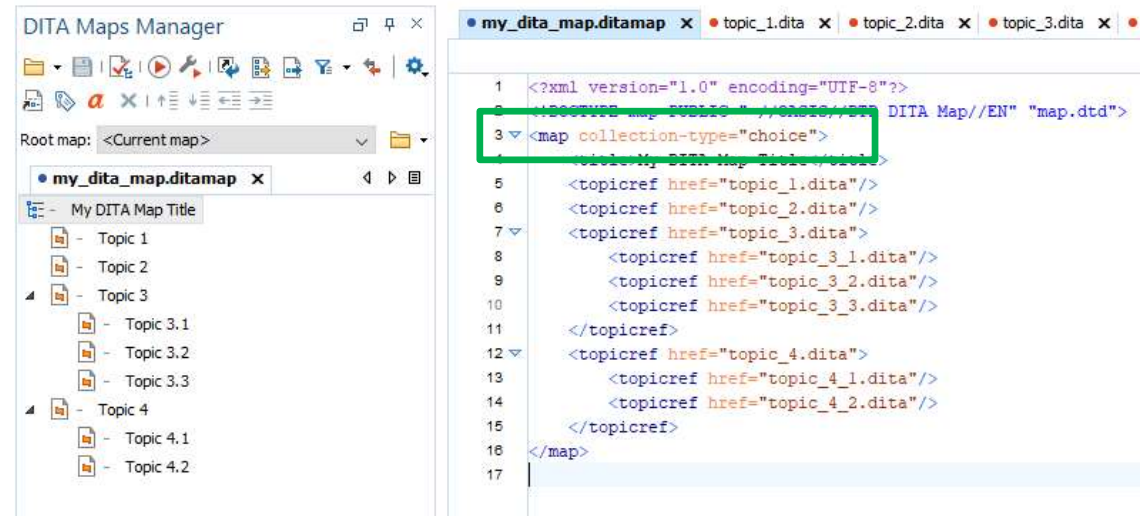
To navigate to this option in Oxygen XML Author:

1. Open the DITA map in the *DITA Maps Manager*.
2. Open the context menu of the DITA Map.
3. In the menu that appears, choose *Edit Properties...*
4. In the dialog that appears, select *Attributes* tab page.
5. Navigate to the *collection-type* attribute.
6. Choose the desired value for this attribute. Let's start with “choice”.
7. Save the file.



Exercise – Step 3

Check if you can see the expected result for your DITA map (view in text mode):



The screenshot displays the DITA Maps Manager interface. On the left, a tree view shows the map structure: 'My DITA Map Title' contains 'Topic 1', 'Topic 2', 'Topic 3', and 'Topic 4'. 'Topic 3' and 'Topic 4' are expanded, showing sub-topics 'Topic 3.1', 'Topic 3.2', 'Topic 3.3' and 'Topic 4.1', 'Topic 4.2' respectively. On the right, the XML code for 'my_dita_map.ditamap' is shown. A green box highlights the opening tag for a choice collection: `<map collection-type="choice">`. The XML code includes topic references for 'topic_1.dita', 'topic_2.dita', 'topic_3.dita' (with sub-topics), and 'topic_4.dita' (with sub-topics).

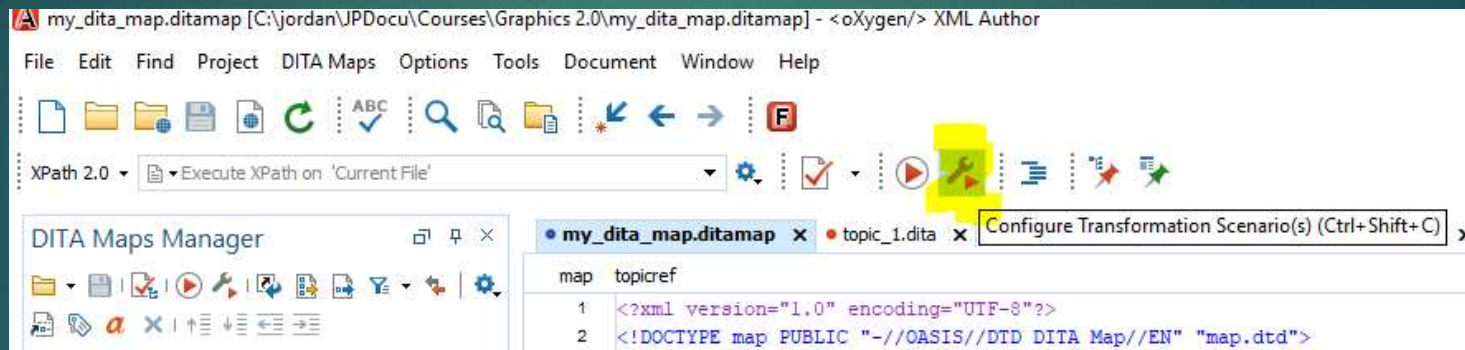
```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE map PUBLIC "-//OASIS//DTD DITA Map//EN" "map.dtd">
3 <map collection-type="choice">
4   <topicref href="topic_1.dita"/>
5   <topicref href="topic_2.dita"/>
6   <topicref href="topic_3.dita">
7     <topicref href="topic_3_1.dita"/>
8     <topicref href="topic_3_2.dita"/>
9     <topicref href="topic_3_3.dita"/>
10  </topicref>
11  <topicref href="topic_4.dita">
12    <topicref href="topic_4_1.dita"/>
13    <topicref href="topic_4_2.dita"/>
14  </topicref>
15 </map>
16
17
```

Reminder: Before Triggering the Transformation

- ▶ To see your links in action, you need to trigger transformation of your content (that is, generate a deliverable).
- ▶ Remember that automatically generated links will have slightly different appearance according to the deliverable type you choose to generate (PDF, HTML and so on.)
- ▶ Watch out for errors, to see if there are problems in your content that require your intervention.
- ▶ Mixing together manually with automatically generated links works, but the results can look ugly (or links can be duplicated accidentally). You need to have a linking strategy defined first!

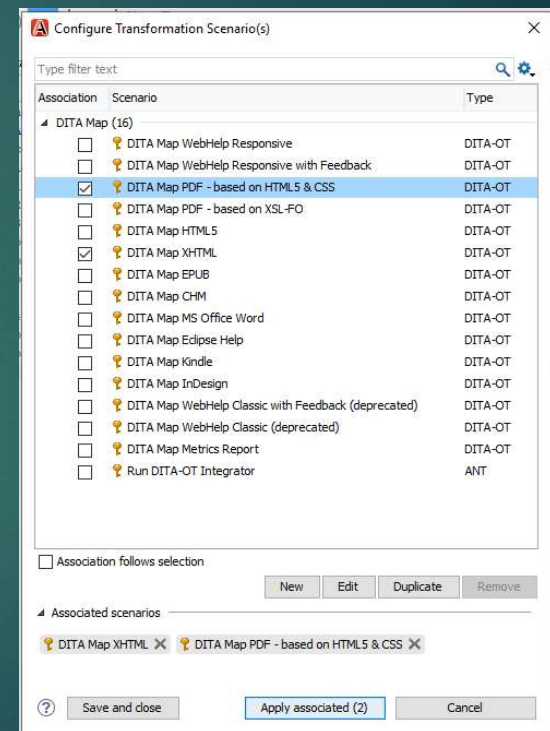
Exercise Step 4 - Trigger Transformation to XHTML

1. Select the DITA map in the DITA Maps Manager.
2. Configure the transformation scenario to apply:



Step 5 – Trigger the Transformation

Choose Apply Associated scenarios.
Wait for the transformation to finish.
Validate the result.



Exercise

Following the steps described so far, generate HTML output with collection-type attribute set to “choice”.

Collection type “family”

- ▶ Automatically generate links to sibling topics (**not** child-parent relations)
- ▶ Suitable for HTML and PDF outputs.



my_dita_map.pdf x

3 (5 of 6)

Chapter 3. Topic 3

<<Text of this topic is here>>

Related concepts

- [Topic 1 \(page 1\)](#)
- [Topic 2 \(page 2\)](#)
- [Topic 4 \(page 4\)](#)

Topic 3.1

<<Text of this topic is here>>

Topic 3.2

<<Text of this topic is here>>

Bookmarks

- My DITA Map Title
- Contents
- Chapter 1. Topic 1
- Chapter 2. Topic 2
- ▼ Chapter 3. Topic 3
 - Topic 3.1
 - Topic 3.2
 - Topic 3.3
- ▼ Chapter 4. Topic 4
 - Topic 4.1
 - Topic 4.2

Exercise

Following the steps described so far, generate HTML output with collection-type attribute set to “family”.

Hint: reuse the content you already have; simply change the attribute value.

Collection type “sequence”

- ▶ Use it to automatically generate next/previous links for online presentation (HTML outputs)



Links to Previous/Next topic:

Topic 3

<<Text of this topic is here>>

[Topic 3.1](#)

[Topic 3.2](#)

[Topic 3.3](#)

Previous topic: [Topic 2](#)

Next topic: [Topic 3.1](#)

Link back to the parent topic:

Topic 3.1

<<Text of this topic is here>>

Parent topic: [Topic 3](#)

Exercise

Following the steps described so far, generate HTML output with collection-type attribute set to “sequence”.

Hint: reuse the content you already have; simply change the attribute value.